RIGHT EXACTNESS OF THE CHOW SEQUENCE OF TWO CLOSED SUBSCHEMES COVERING A SCHEME

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with \input{preamble} and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-chow/preamble.tex is excerpted verbatim as three spans, lines 8-12 (the verbatim package together with the project's own reference, slogan and history display-environment definitions, including the author's commented-out older form of the reference environment), lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). The lines 8-12 span is required because a counted statement or its surrounding text may use those display environments, which the project typesets as comments. Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- Surjectivity of the sum of the two pushforwards. Same chapter chow.tex, label lemma-exact-sequence-closed, invoked at proof line 3009. Gives the surjectivity of the arrow from the direct sum to the Chow group of the union, which is half of the exactness.
- Definition of rational equivalence as a locally finite sum of principal divisors. Same chapter chow.tex, definition used at proof lines 3011-3019. Lets a sum of two cycle classes mapping to zero be written as a locally finite sum of divisors of rational functions on integral closed subschemes of the union.

Source reference. Original lemma 0F92: sources/stacks-chow/pinned-original-source.tex, source lines 1964–1964. Navigation pointer for \ref{lemma-exact-sequence-closed} used inside the counted span (line 3009); the argument that consumes it is retained in the span.

Source reference. Original reference 02QL: sources/stacks-chow/pinned-original-source.tex, source lines 1372–1372. Navigation pointer for \ref{situation-setup} used inside the counted span (line 2992); the argument that consumes it is retained in the span.

Lemma 0.1. *Let (S, δ) be as in Situation 02QL. Let X be locally of finite type over S. Let $X_1, X_2 \subset X$ be closed subschemes such that $X = X_1 \cup X_2$ set theoretically.*

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For every $k \in \mathbf{Z}$ the sequence of abelian groups

$$\mathrm{CH}_k(X_1 \cap X_2) \longrightarrow \mathrm{CH}_k(X_1) \oplus \mathrm{CH}_k(X_2) \longrightarrow \mathrm{CH}_k(X) \longrightarrow 0$$

is exact. Here $X_1 \cap X_2$ is the scheme theoretic intersection and the maps are the pushforward maps with one multiplied by -1.

Proof. By Lemma 0F92 the arrow $\mathrm{CH}_k(X_1) \oplus \mathrm{CH}_k(X_2) \to \mathrm{CH}_k(X)$ is surjective. Suppose that (α_1, α_2) maps to zero under this map. Write $\alpha_1 = \sum n_{1,i}[W_{1,i}]$ and $\alpha_2 = \sum n_{2,i}[W_{2,i}]$. Then we obtain a locally finite collection $\{W_j\}_{j \in J}$ of integral closed subschemes of X of δ-dimension $k+1$ and $f_j \in R(W_j)^*$ such that

$$\sum n_{1,i}[W_{1,i}] + \sum n_{2,i}[W_{2,i}] = \sum (i_j)_* \mathrm{div}(f_j)$$

as cycles on X where $i_j : W_j \to X$ is the inclusion morphism. Choose a disjoint union decomposition $J = J_1 \amalg J_2$ such that $W_j \subset X_1$ if $j \in J_1$ and $W_j \subset X_2$ if $j \in J_2$. (This is possible because the W_j are integral.) Then we can write the equation above as

$$\sum n_{1,i}[W_{1,i}] - \sum\nolimits_{j \in J_1} (i_j)_* \mathrm{div}(f_j) = -\sum n_{2,i}[W_{2,i}] + \sum\nolimits_{j \in J_2} (i_j)_* \mathrm{div}(f_j)$$

Hence this expression is a cycle (!) on $X_1 \cap X_2$. In other words the element (α_1, α_2) is in the image of the first arrow and the proof is complete. $\square$